



Inspiring Excellence

Data Modeling

Data Flow Diagram

Models: Logical and Physical

Model – a pictorial representation of reality.

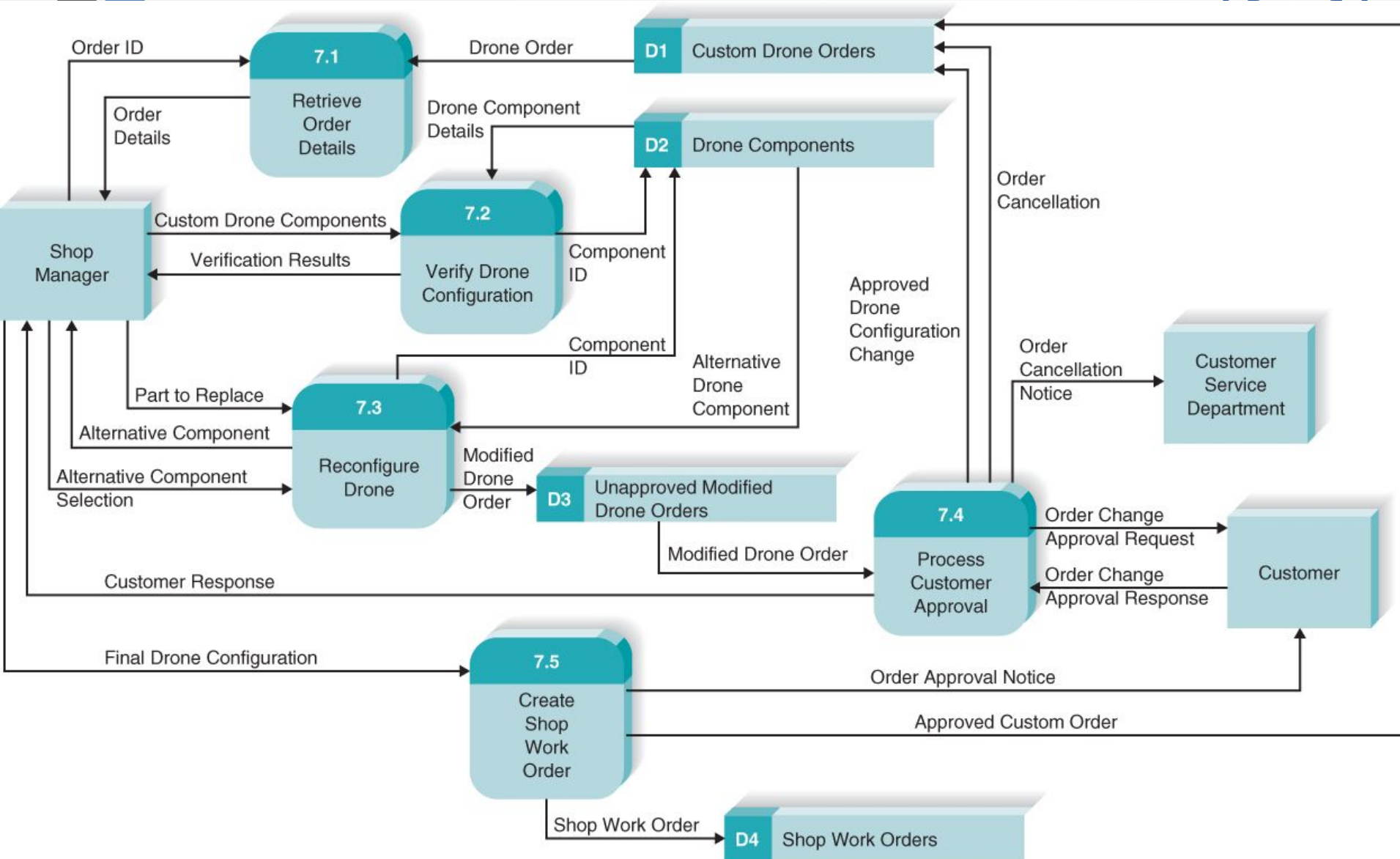
Just as a picture is worth a thousand words, most models are pictorial representations of reality.

Logical model – a nontechnical pictorial representation that depicts what a system is or does. Synonyms or *essential model*, *conceptual model*, and *business model*.

Physical model – a technical pictorial representation that depicts what a system is or does and how the system is implemented. Synonyms are *implementation model* and *technical model*.

What is a Data Flow Diagram?

- Data flow diagramming is a technique that diagrams the business processes and the data that pass among them
- We use data flow diagrams (DFDs) to describe the to-be system's interactions with its environment, processes, flows of data, and data stores
- Logical process models are models that describe processes, without suggesting how they are conducted
- Physical models provide information that is needed to ultimately build the system



Process

Process – work performed by a system in response to incoming data flows or conditions. A synonym is *transform*.

- All information systems include processes - usually many of them
- Processes respond to business events and conditions and transform data into useful information
- Modeling processes helps us to understand the interactions with the system's environment, other systems, and other processes.
- Named with a strong action verb followed by object clause describing what the work is performed on/for .

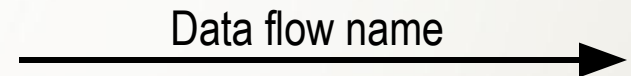


Gane and Sarson shape

Data Flows & Control Flows

Data flow – data that is input to or output from a process.

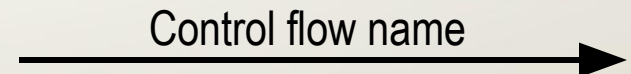
- A data flow is data in motion
- A data flow may also be used to represent the creation, reading, deletion, or updating of data in a file or database (called a data store).



Composite data flow – a data flow that consists of other data flows.

Control flow – a condition or nondata event that triggers a process.

- Used sparingly on DFDs.



Data Stores

Data store – stored data intended for later use.
Synonyms are *file* and *database*.

- Frequently implemented as a file or database.
- A data store is “data at rest” compared to a data flow that is “data in motion.”
- Almost always one of the following:
 - Persons, Places, Objects etc
- Data stores depicted on a DFD store all instances of data entities (depicted on an ERD)



Gane and Sarson shape



DeMarco/Yourdon shape

External Agents

External agent – an outside person, unit, system, or organization that interacts with a system. Also called an *external entity*.

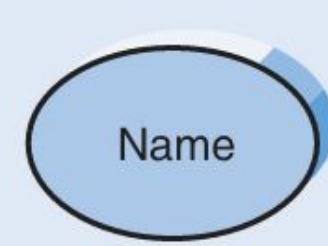
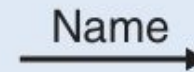
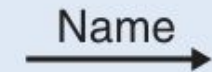
- External agents define the “boundary” or scope of a system being modeled.
- As scope changes, external agents can become processes, and vice versa.



Gane and Sarson shape

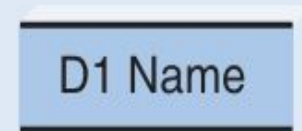
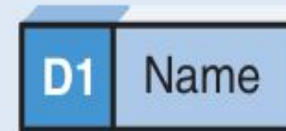


DeMarco/Yourdon shape

Data Flow Diagram Element	Typical Computer-Aided Software Engineering Fields	Gane and Sarson Symbol	DeMarco and Yourdon Symbol
Every <i>process</i> has - a number - a name (verb phrase) - a description - at least one output data flow - at least one input data flow	Label (name) Type (process) Description (what is it) Process number Process description (structured English) Notes		
Every <i>data flow</i> has - a name (a noun) - a description - one or more connections to a process	Label (name) Type (flow) Description Alias (another name) Composition (description of data elements) Notes		

Every *data store* has
a number
a name (a noun)
a description
one or more input
data flows
one or more output
data flows

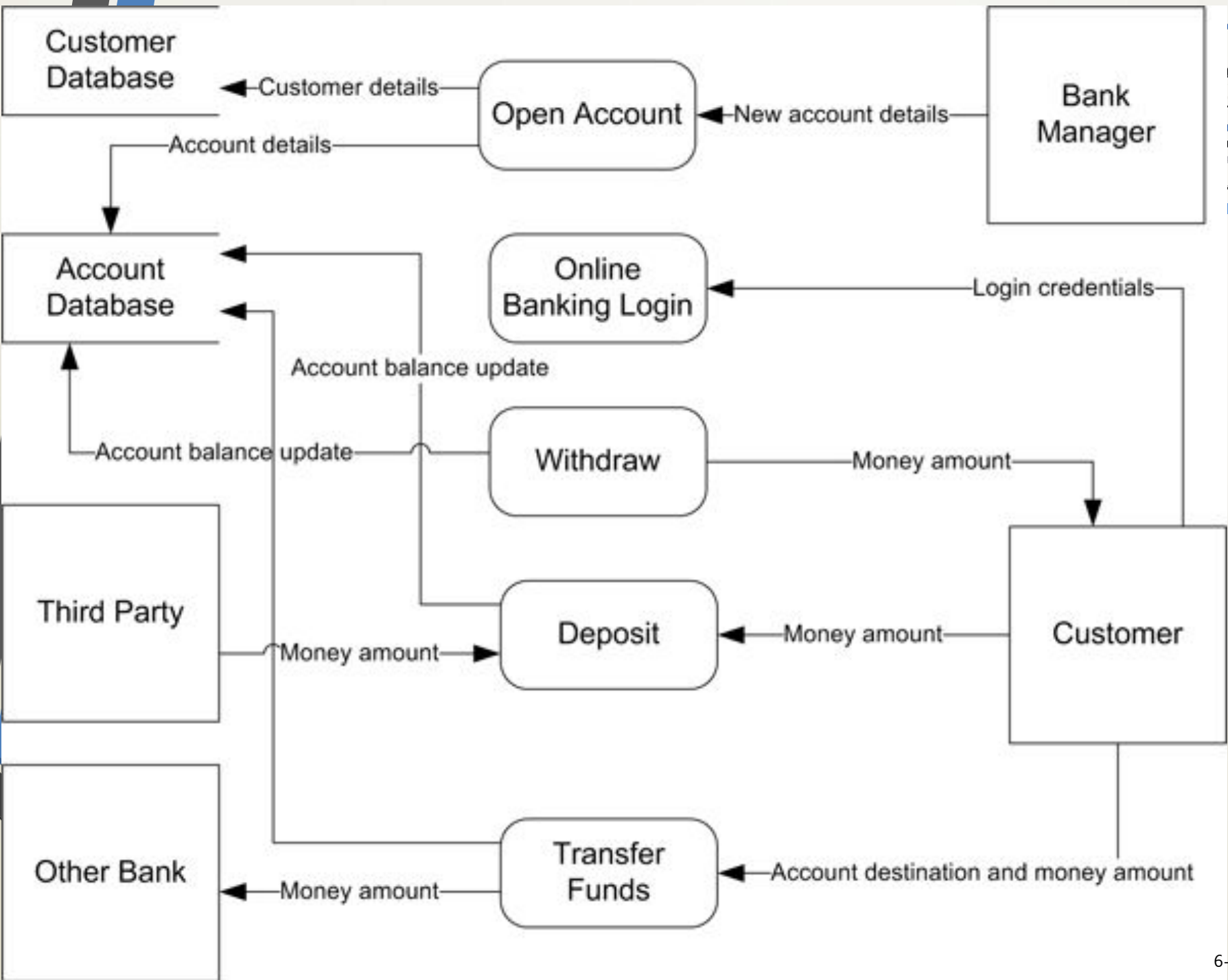
Label (name)
Type (store)
Description
Alias (another name)
Composition
(description of data
elements)
Notes



Every *external entity* has
a name (a noun)
a description

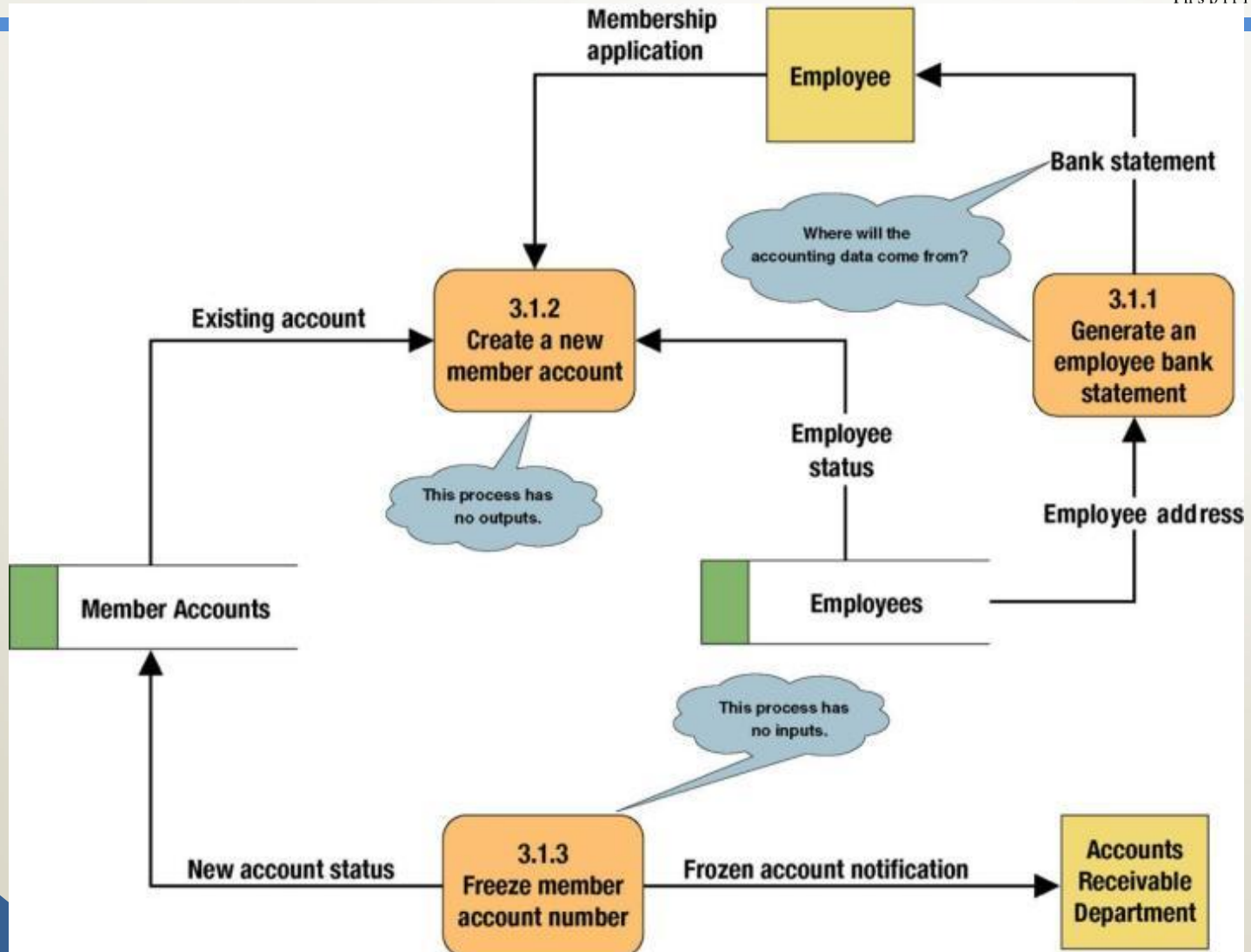
Label (name)
Type (entity)
Description
Alias (another name)
Entity description
Notes





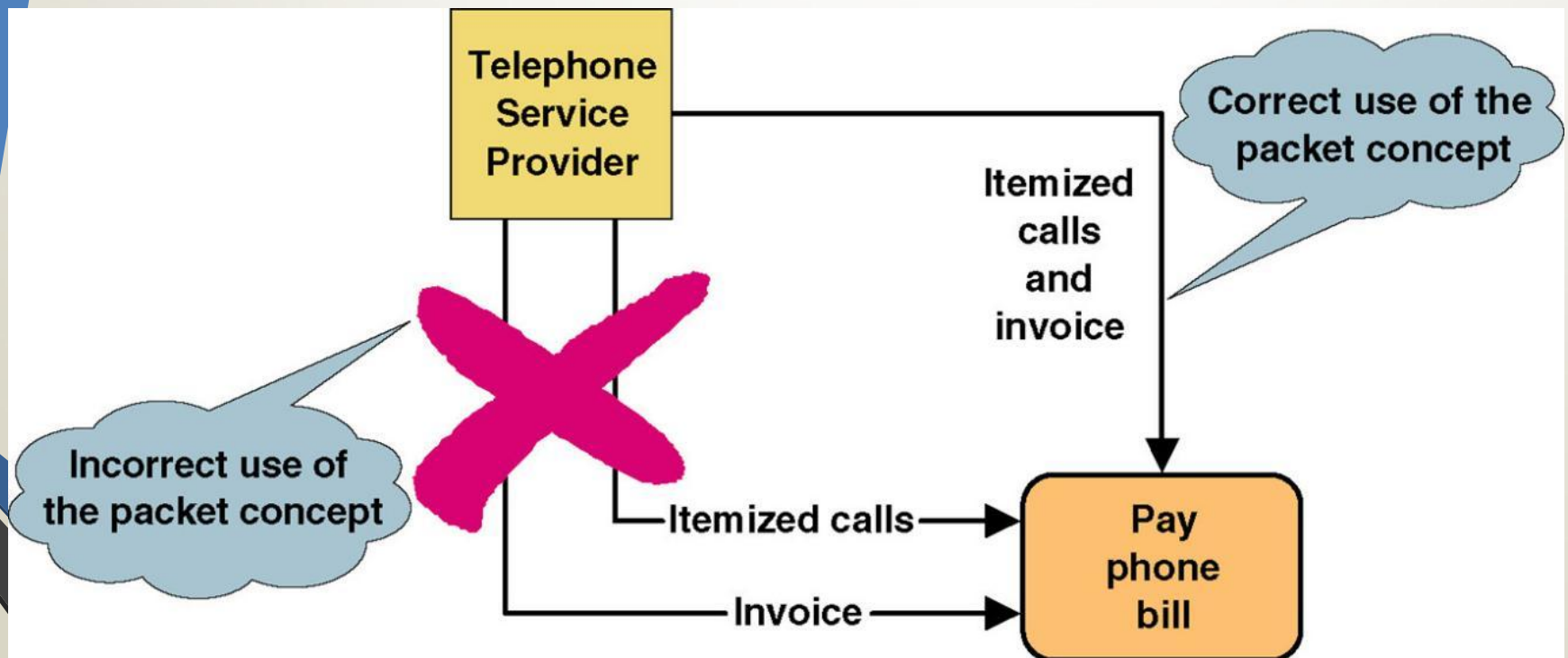
Common Process Errors on DFDs

Common Process Errors on DFDs

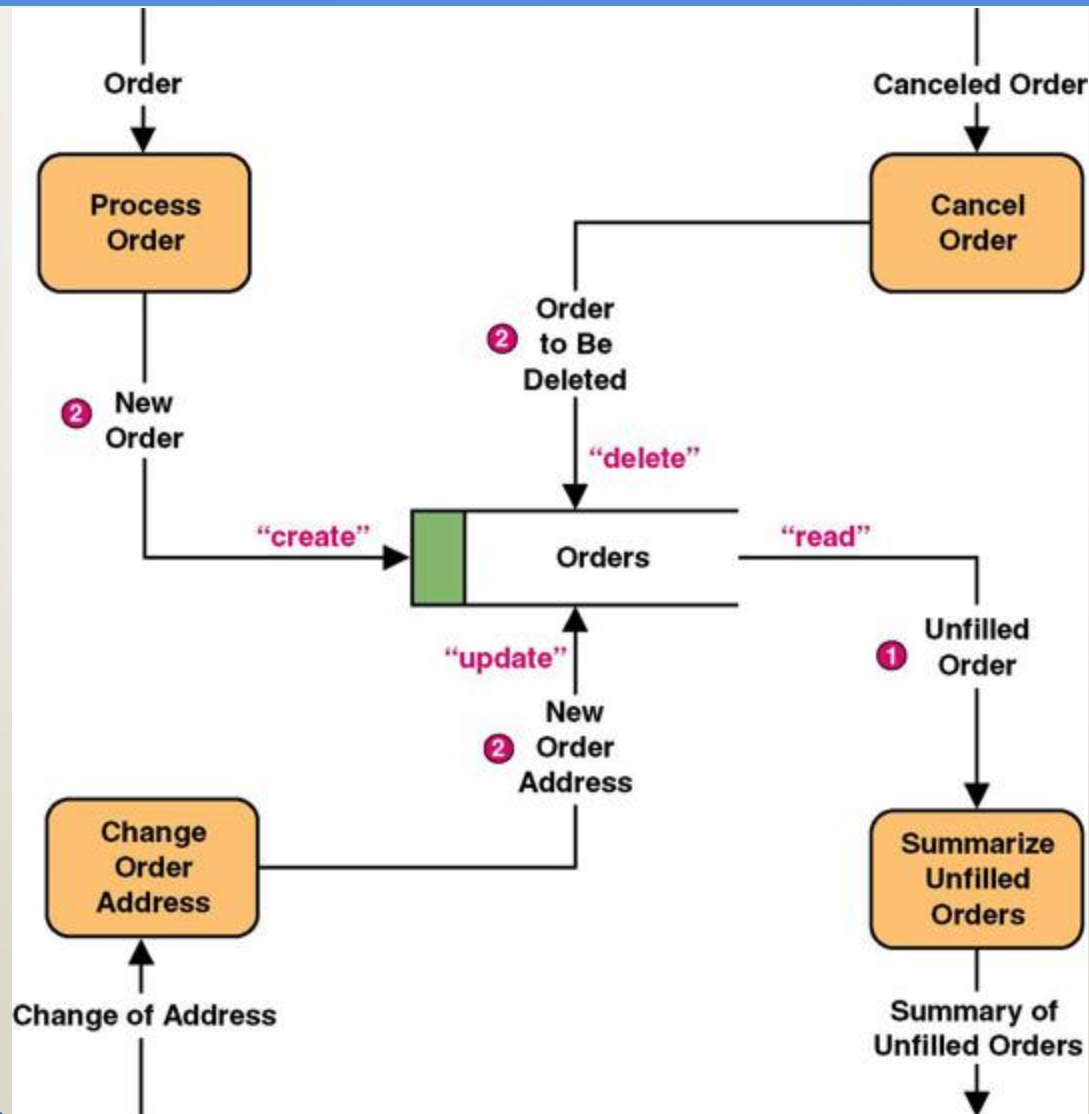


Data Flow Packet Concept

- Data that should travel together should be shown as a single data flow, no matter how many physical documents might be included.



Data Flows to and from Data Stores

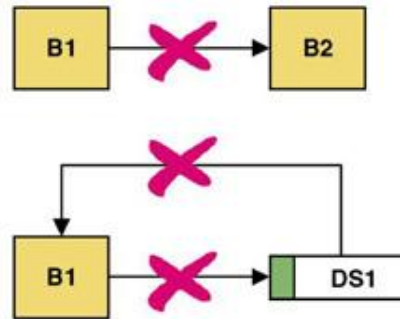


Rules for Data Flows

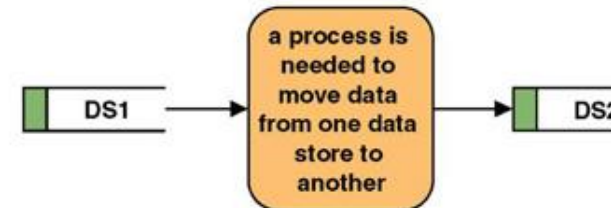
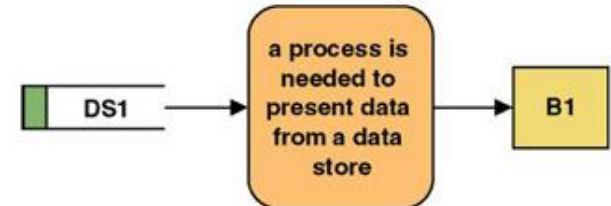
- A data flow should never go unnamed.
- In logical modeling, data flow names should describe the data flow without describing the implementation
- All data flows must begin and/or end at a process.



Illegal
data
flows



Corrected
data
flows



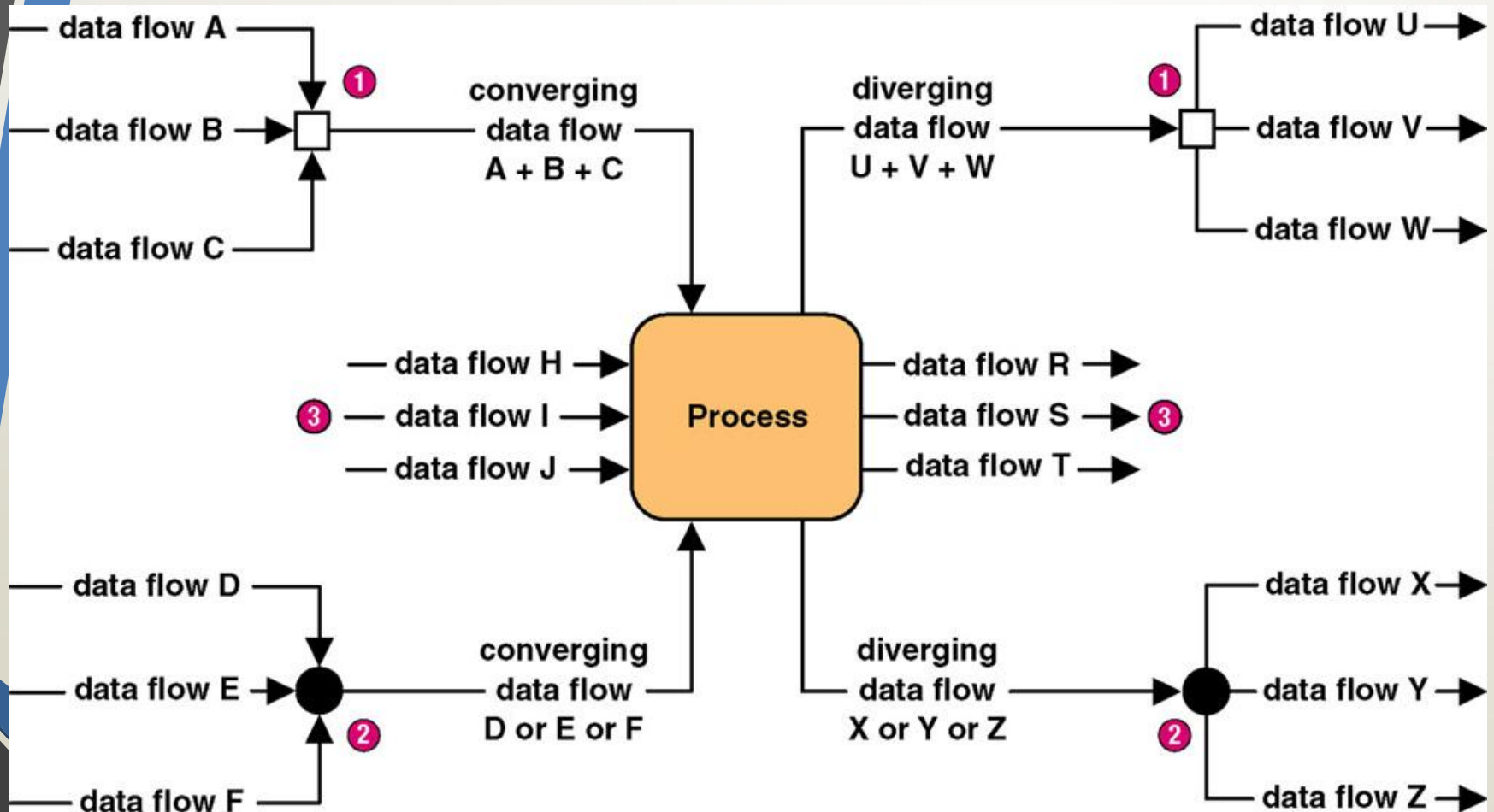
Diverging and Converging Data Flows



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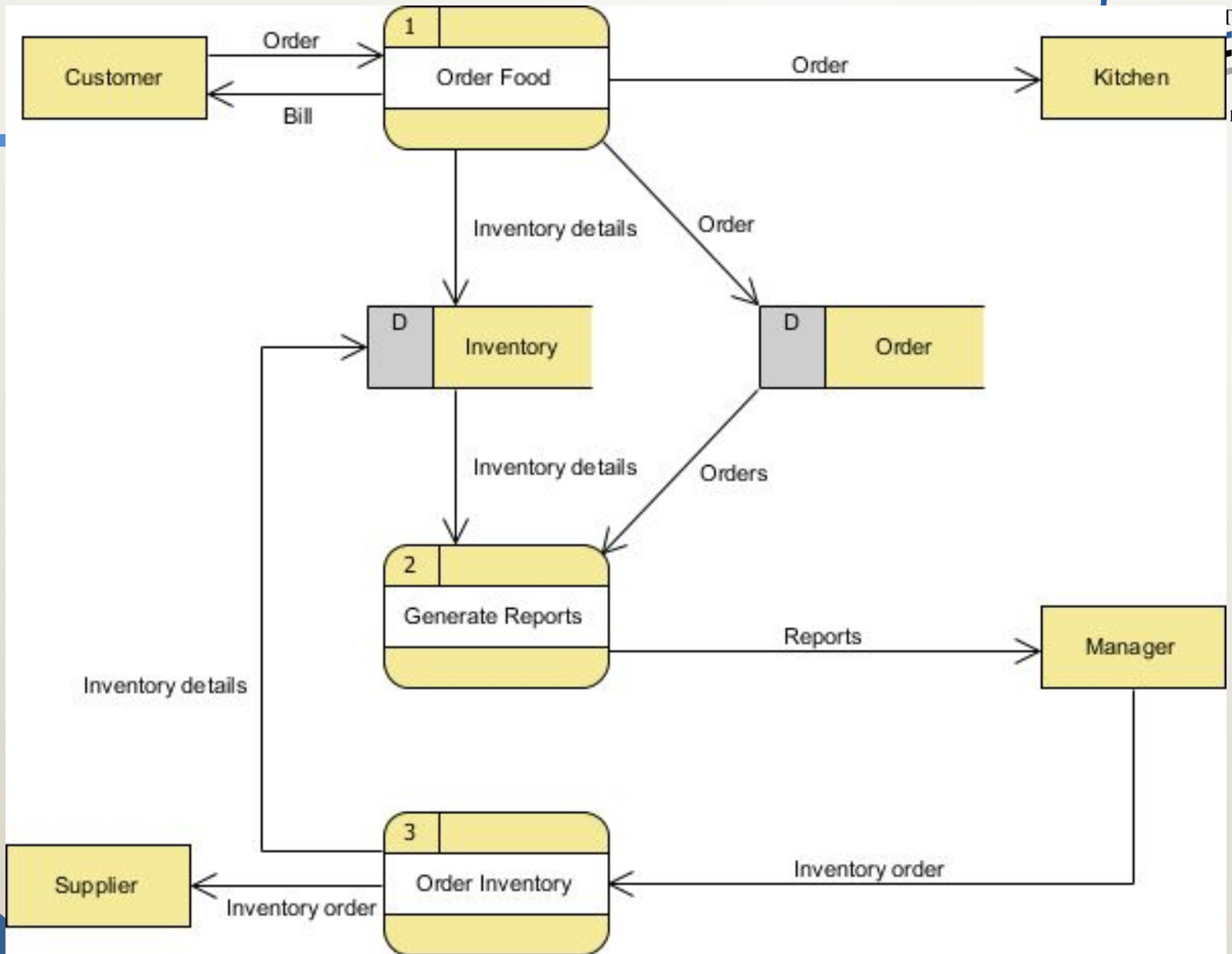
- **Diverging data flow** – a data flow that splits into multiple data flows.
 - Indicates data that starts out naturally as one flow, but is routed to different destinations.
 - Also useful to indicate multiple copies of the same output going to different destinations.
- **Converging data flow** – the merger of multiple data flows into a single packet.
 - Indicates data from multiple sources that can (must) come together as a single packet for subsequent processing.

Diverging and Converging Data Flows

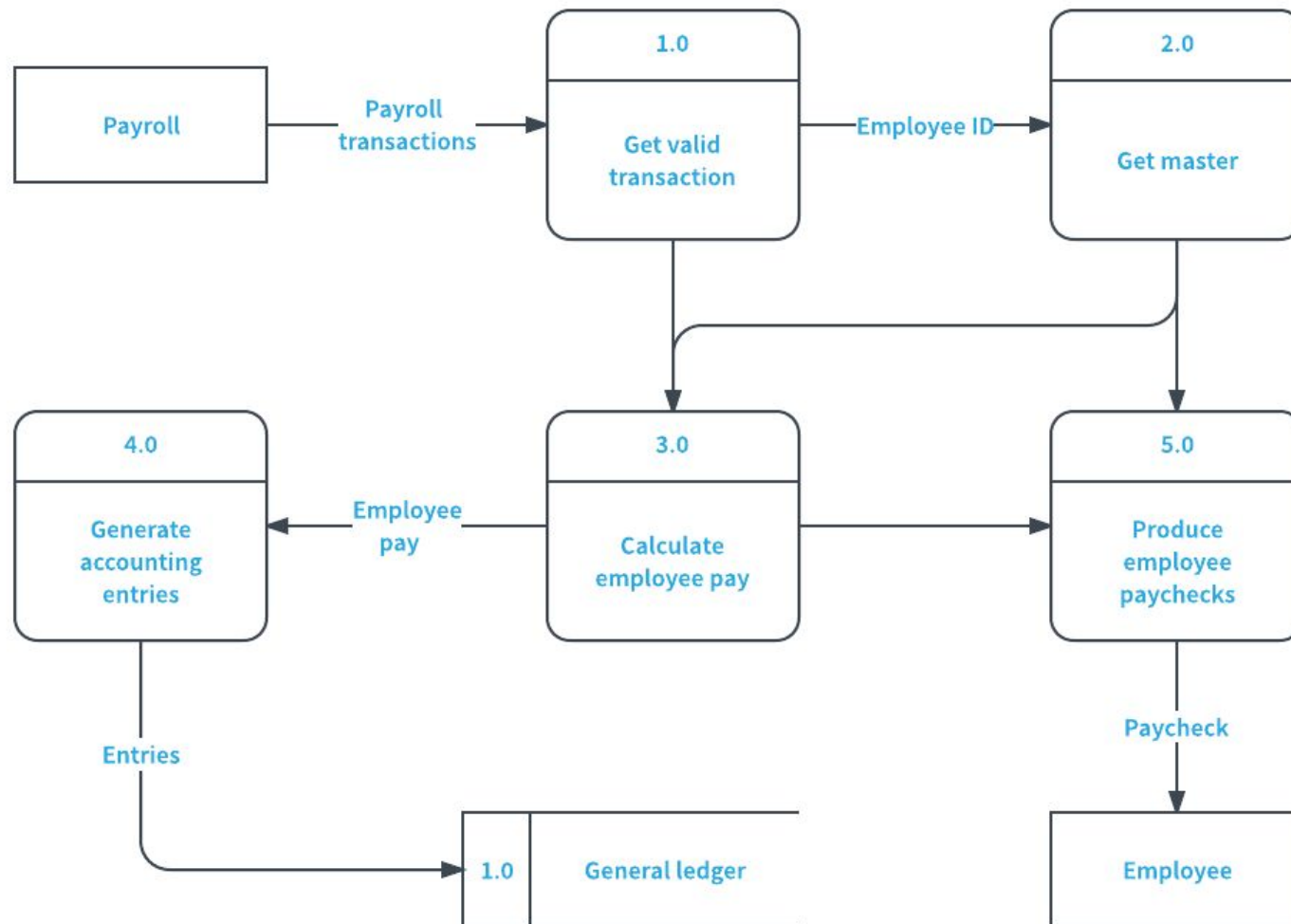


Exercise: Order food

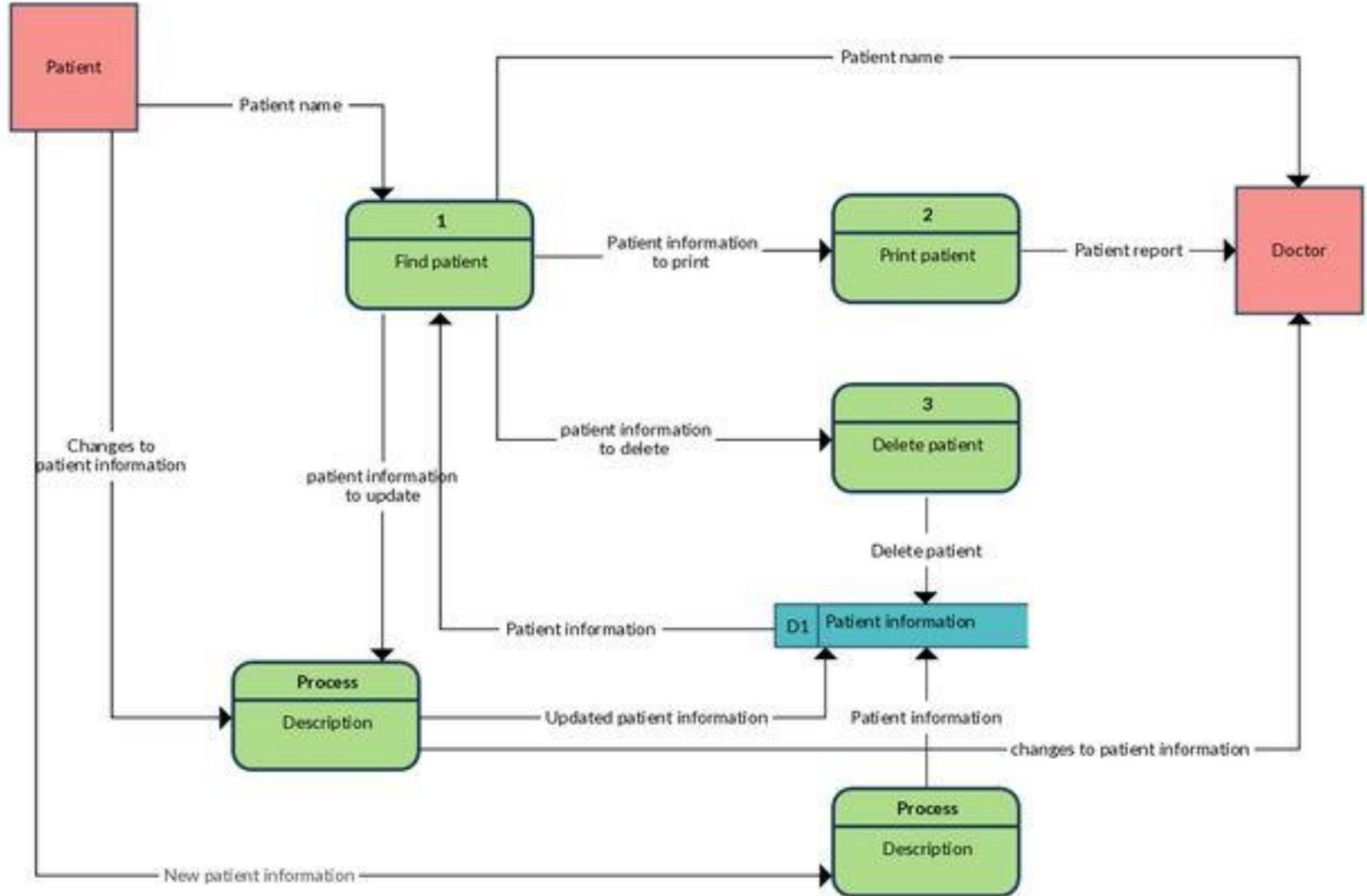
- Draw a Data Flow Diagram (DFD) for “***Food Ordering System***” considering the following scenarios:
- **Customer** place an *Order*. The Order Food process receives the *Order*, forwards it to the **Kitchen**, store it in the ***Order*** data store, and store the updated *Inventory details* in the ***Inventory*** data store. The process also delivers a *Bill* to the **Customer**. **Manager** can receive *Reports* through the Generate Reports process, which takes *Inventory details* and *Orders* as input from the ***Inventory*** and ***Order*** data store respectively. **Manager** can also initiate the Order Inventory process by providing *Inventory order*. The process forwards the *Inventory order* to the **Supplier** and stores the updated *Inventory details* in the ***Inventory*** data store.

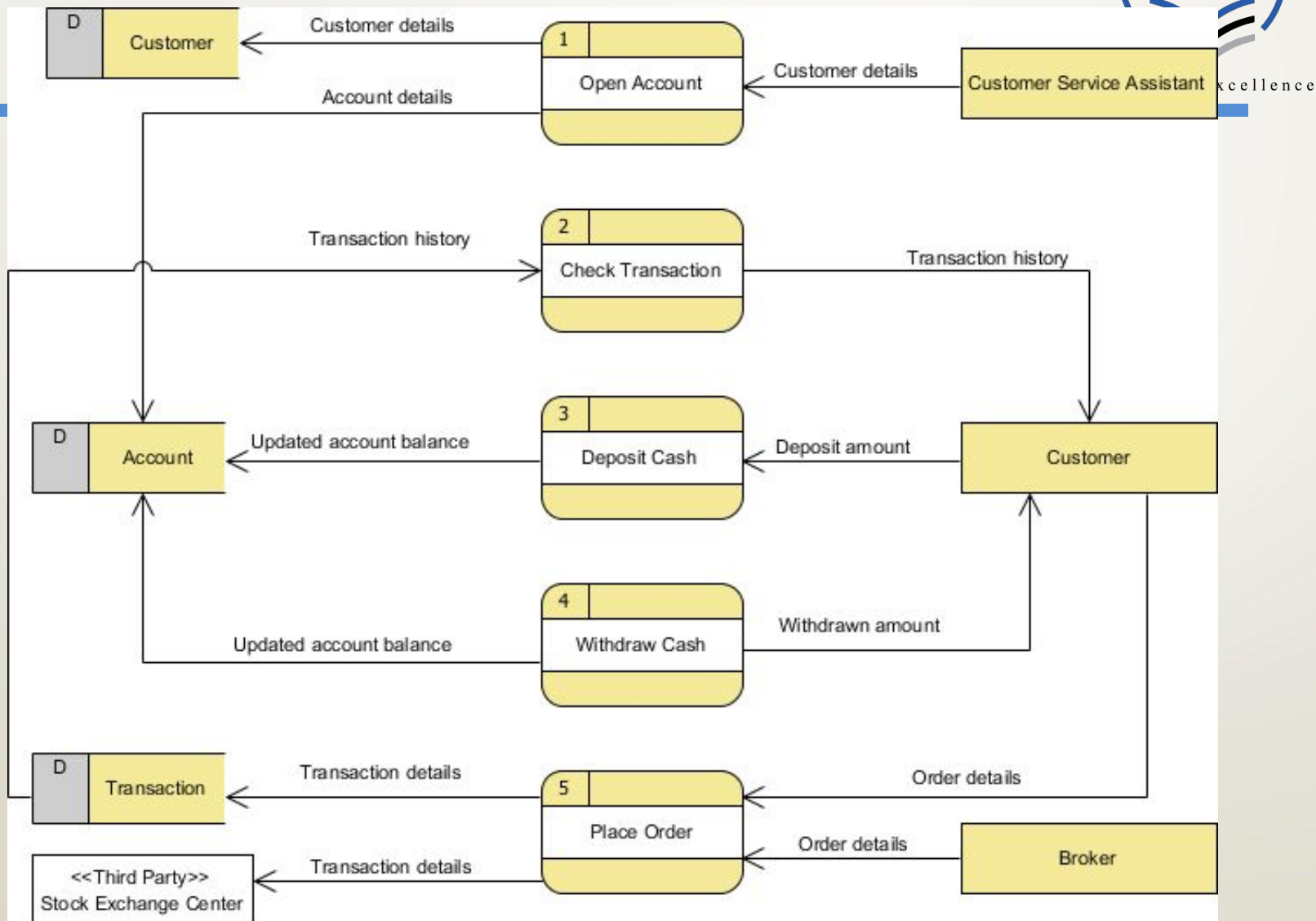


Payroll management system



Patient Information system

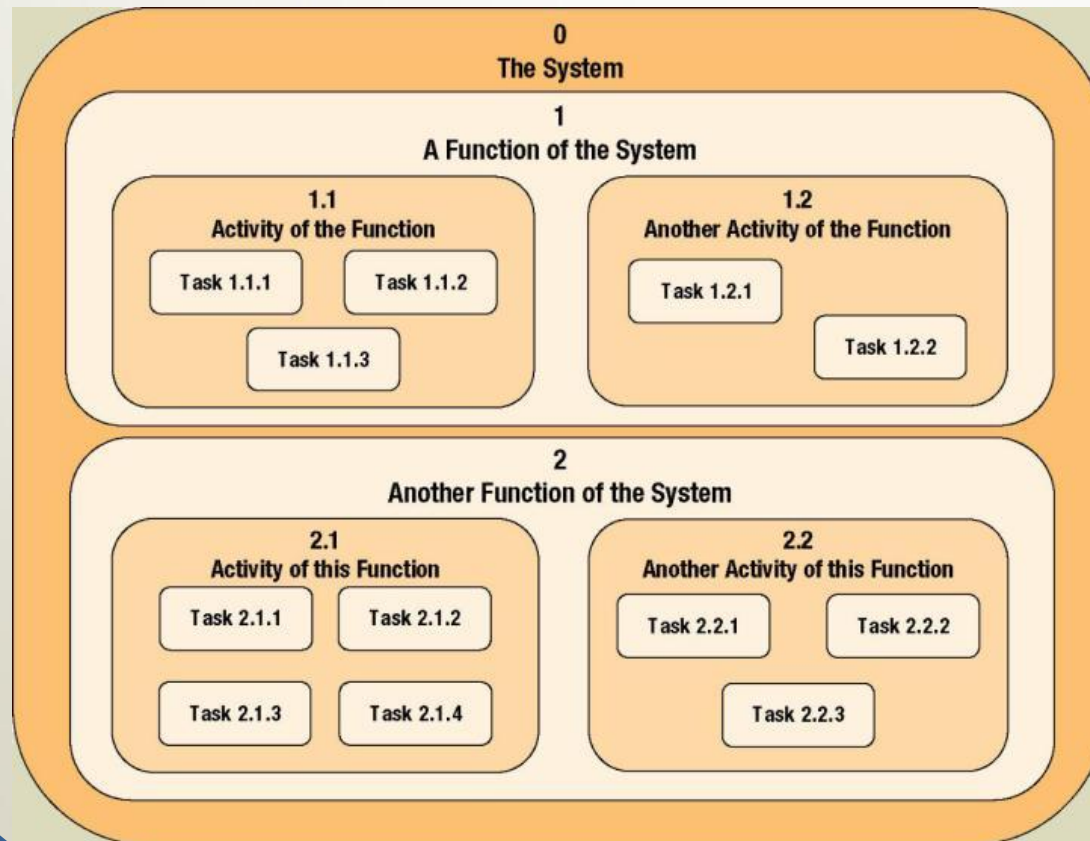




Process Decomposition

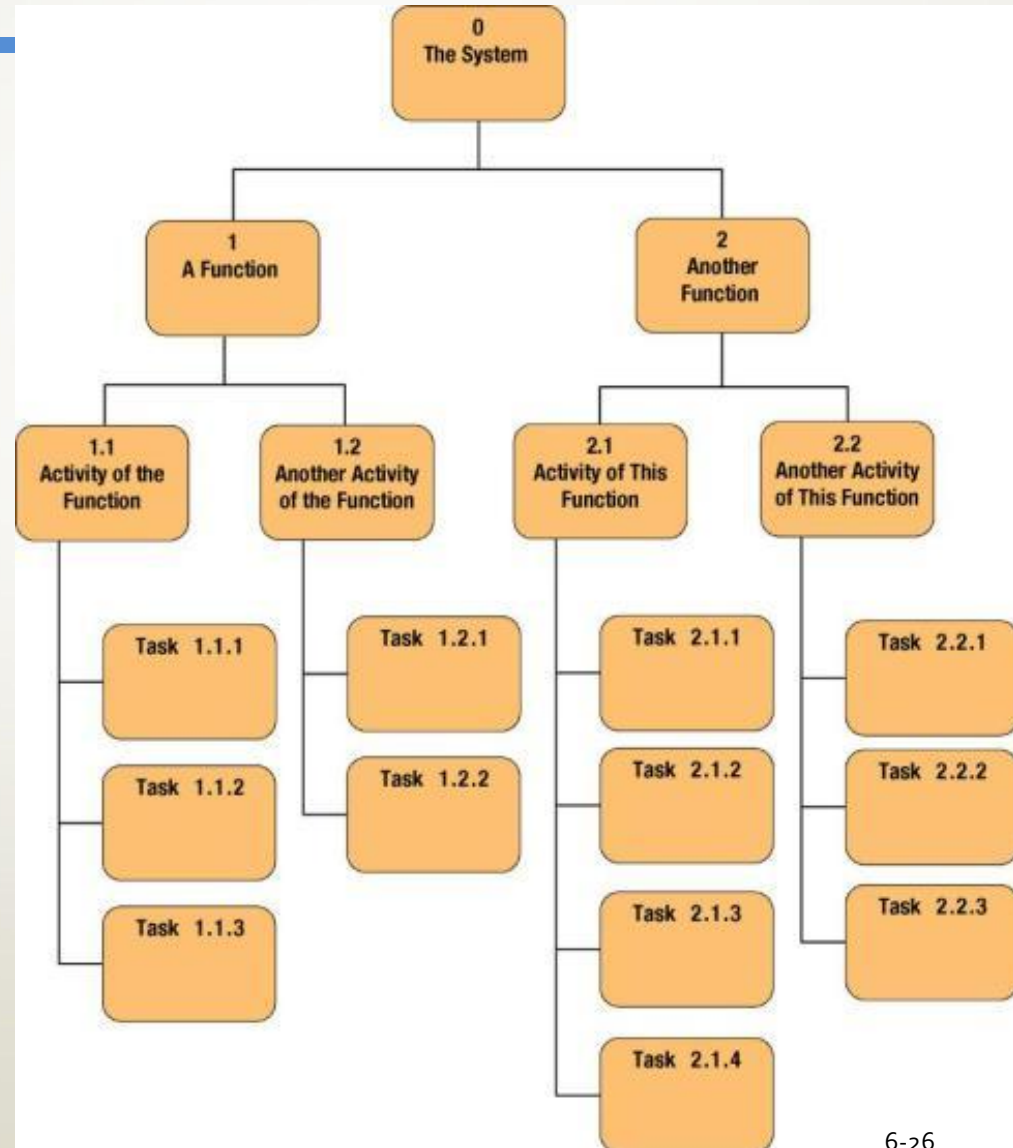
Process Decomposition

Decomposition – the act of breaking a system into sub-components. Each level of abstraction reveals more or less detail.



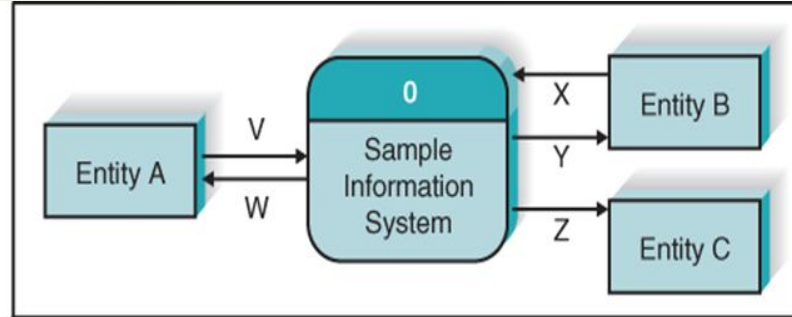
Decomposition Diagrams

Decomposition diagram – a tool used to depict the decomposition of a system. Also called hierarchy chart.

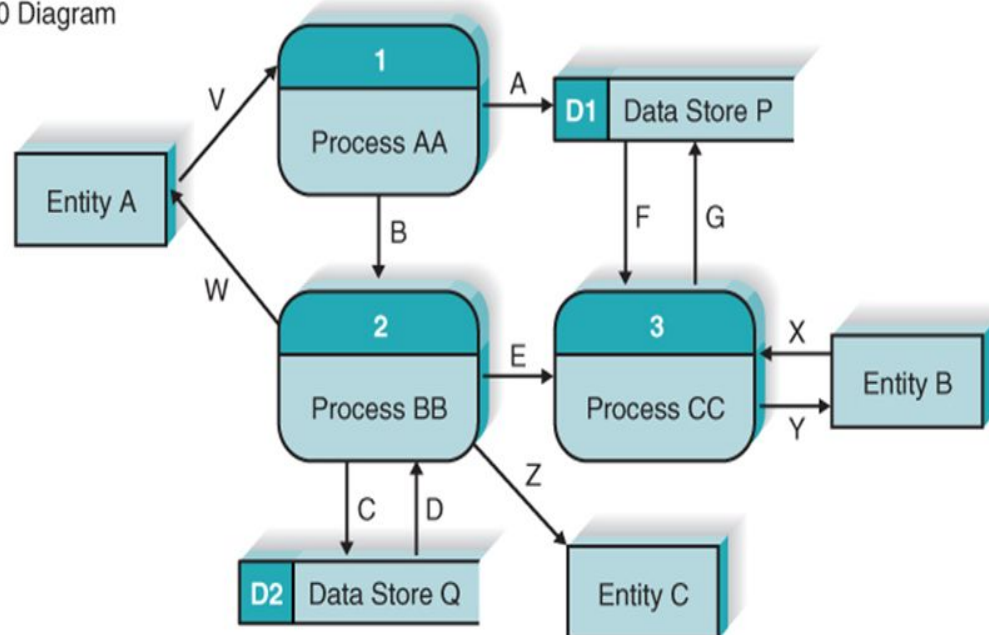


DFD Hierarchy (1 of 3)

Context Diagram



Level 0 Diagram

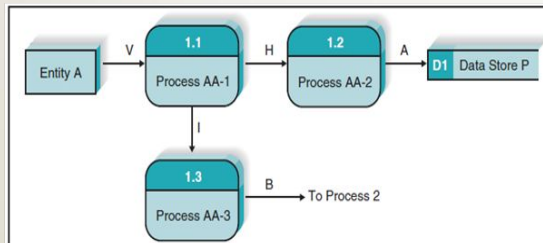
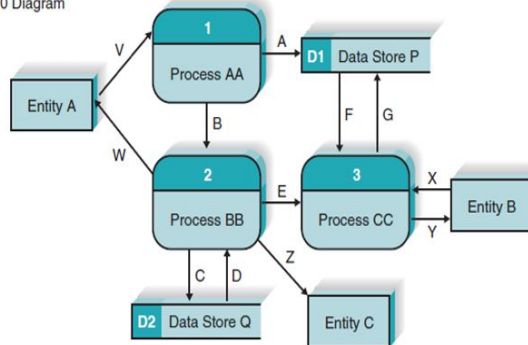


DFD Hierarchy (2 of 3)

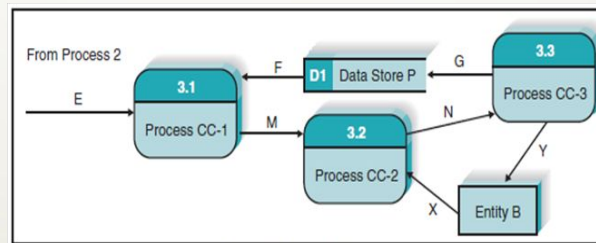
- Processes on Level 0 diagram each decompose into separate Level 1 diagrams
- Processes on Level 1 diagrams may or may not be decomposed into separate Level 2 diagrams
- Processes are decomposed until each process is a single-purpose, primitive process

DFD Hierarchy (3 of 3)

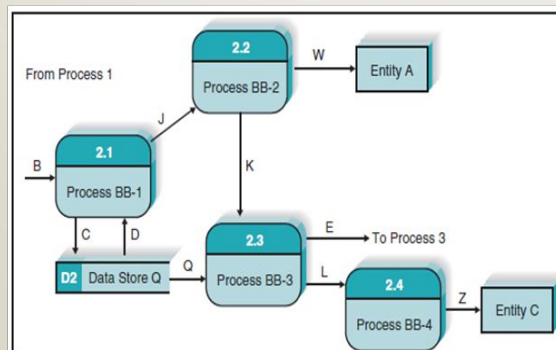
Level 0 Diagram



Process 1 Level 1 Diagram



Process 3 Level 1 Diagram



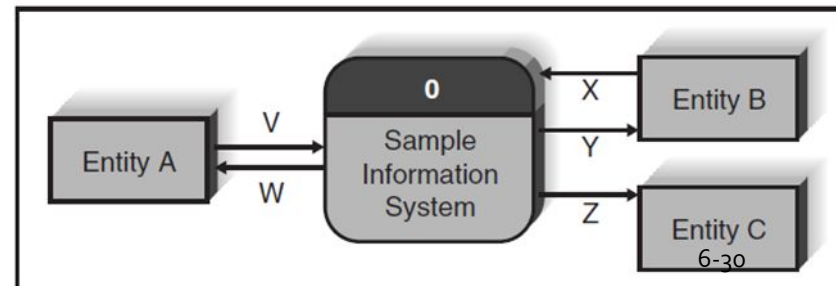
Process 2 Level 1 Diagram

Context Data Flow Diagram

Context data flow diagram - a process model used to document the scope for a system. Also called the environmental model.

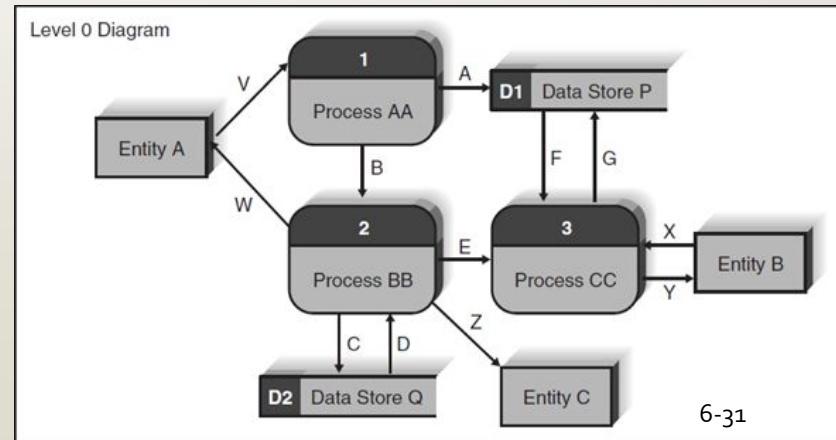
- Top-level DFD in every process model
- Think of the system as a "black box."
- Ask users what business transactions the system must respond to. These are inputs, and the sources are external agents.
- Ask users what responses must be produced by the system. These are outputs, and the destinations are external agents.
- Identify any external data stores, if any.

Context
Diagram



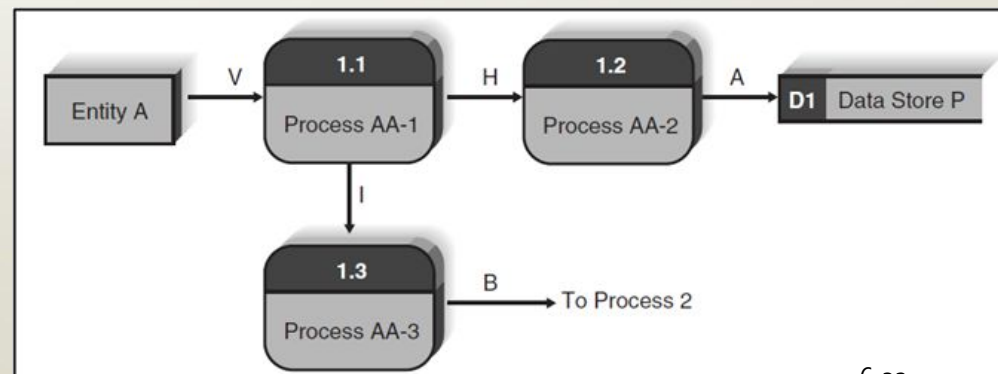
Level 0 Diagram

- Shows all the major processes that comprise the overall system
 - Internal components of process 0
- Shows how the major processes are interrelated by data flows
- Shows external entities and the major processes with which they interact
- Adds stored data via the data stores



Level 1 Diagrams

- Create one level 1 diagram for every major process on level 0 diagram
- Shows the internal processes that comprise a single process on the level 0 diagram
- Shows how information moves to and from each of these processes
- If a parent process is decomposed into, say, three child processes, then these three child processes wholly and completely make up the parent process



Process 1 Level 1 Diagram

Level 2 Diagrams

- Shows all processes that comprise a single process on the level 1 diagram
- Shows how information moves from and to each of these processes
- Level 2 diagrams may not be needed for all level 1 processes
- Correctly numbering each process helps the user understand where the process fits into the overall system

Diagram Numbering

- Correctly numbering each process helps the user understand where the process fits into the overall hierarchy
- Context Diagram is always “Process 0”
- Level 0 processes are always numbered with integer value (1, 2, 3, and so on)
- Level 1 processes always have one “dot”: parent number “dot” unique number (1.1, 1.2, 1.3, and so on)
- Level 2 processes always have two “dots”: parent number “dot” unique number (1.1.1, 1.1.2, 1.1.3, and so on)

Balancing

Balancing - a concept that requires that data flow diagrams at different levels of detail reflect consistency and completeness

- Quality assurance technique
- Requires that if you explode a process to another DFD to reveal more detail, you must include the same data flows and data stores

Validating the Data Flow Diagrams



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Process	• Every process has a unique name that is an action-oriented verb phrase, a number, and a description. • Every process has at least one input data flow. • Every process has at least one output data flow. • Output data flows usually have different names than input data flows because the process changes the input into a different output in some way. • There are between three and seven processes per DFD.
Data Flow	• Every data flow has a unique name that is a noun, and a description. • Every data flow connects to at least one process. • Data flows only in one direction (no two-headed arrows). • A minimum number of data flow lines cross.
Data Store	• Every data store has a unique name that is a noun, and a description. • Every data store has at least one input data flow (which means to add new data or change existing data in the data store) on some page of the process model. • Every data store has at least one output data flow (which means to read data from the data store) on some page of the process model.
External Entity	• Every external entity has a unique name that is a noun, and a description. • Every external entity has at least one input or output data flow.

Creating Data Flow Diagrams (1 of 2)



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- Build the context diagram
 - Identify the external entities and the major inflows they supply and the outflows they receive
- Identify all major processes encompassed by the Context Diagram
 - Each major event / use case is “handled” by a process
- Create DFD “fragments” for each event / use case
 - Each DFD fragment is a mini-diagram showing the process and the external entities and data stores with which it interacts

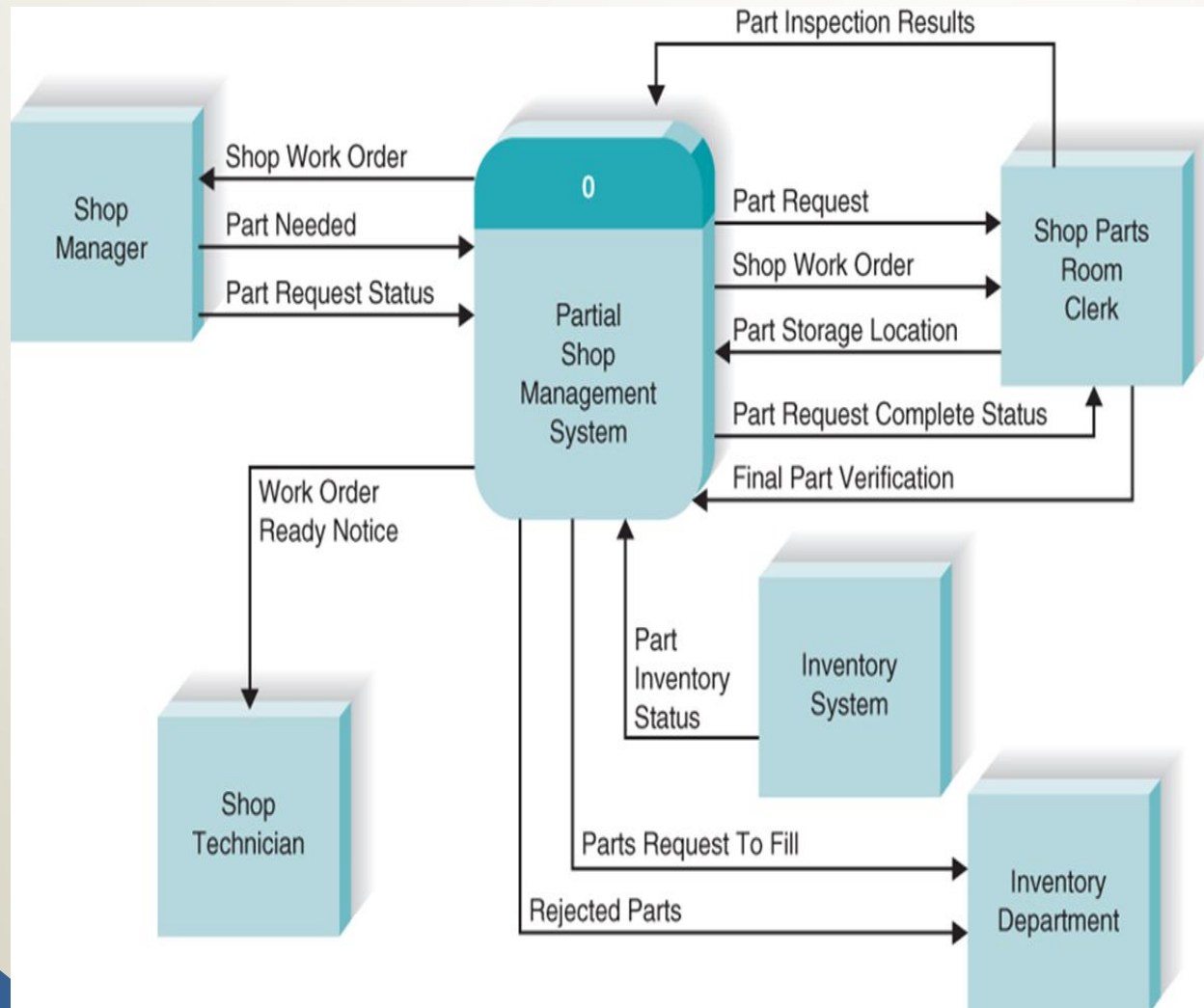
Creating Data Flow Diagrams (2 of 2)

- Organize DFD fragments into level 0 diagram
- Decompose each level 0 process into a level 1 diagram; decompose level 1 processes into level 2 diagrams as needed; etc.
- Validate DFDs with user to ensure completeness and correctness

Illustration – Developing DFDs

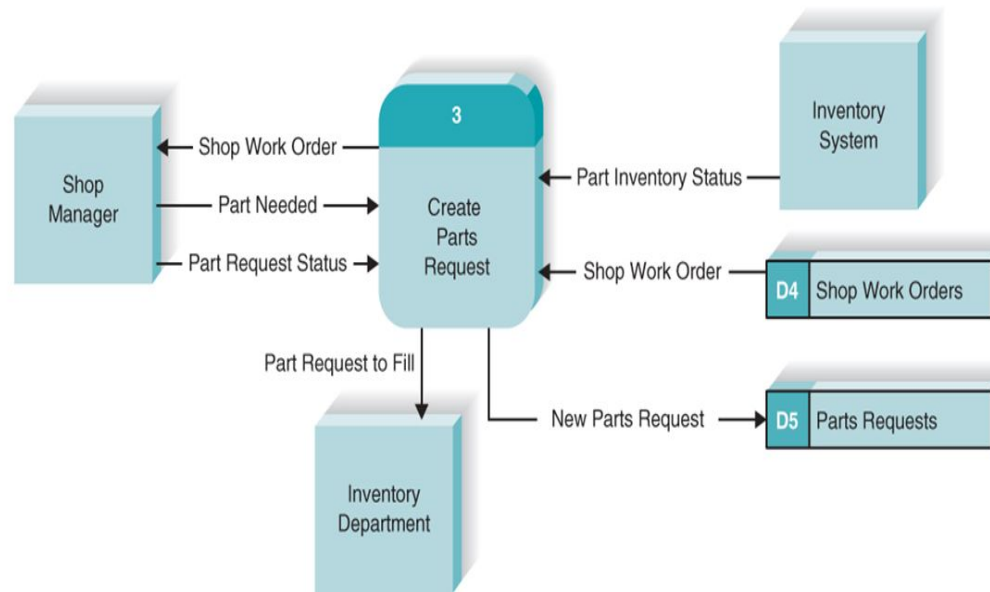
1. Creating the context diagram
2. Create a DFD “fragment” based on a use case
3. Merge DFD “fragment” diagrams into the Level 0 diagram
4. Develop level 1 diagrams for every process on the level 0 diagram

Creating the Context Diagram

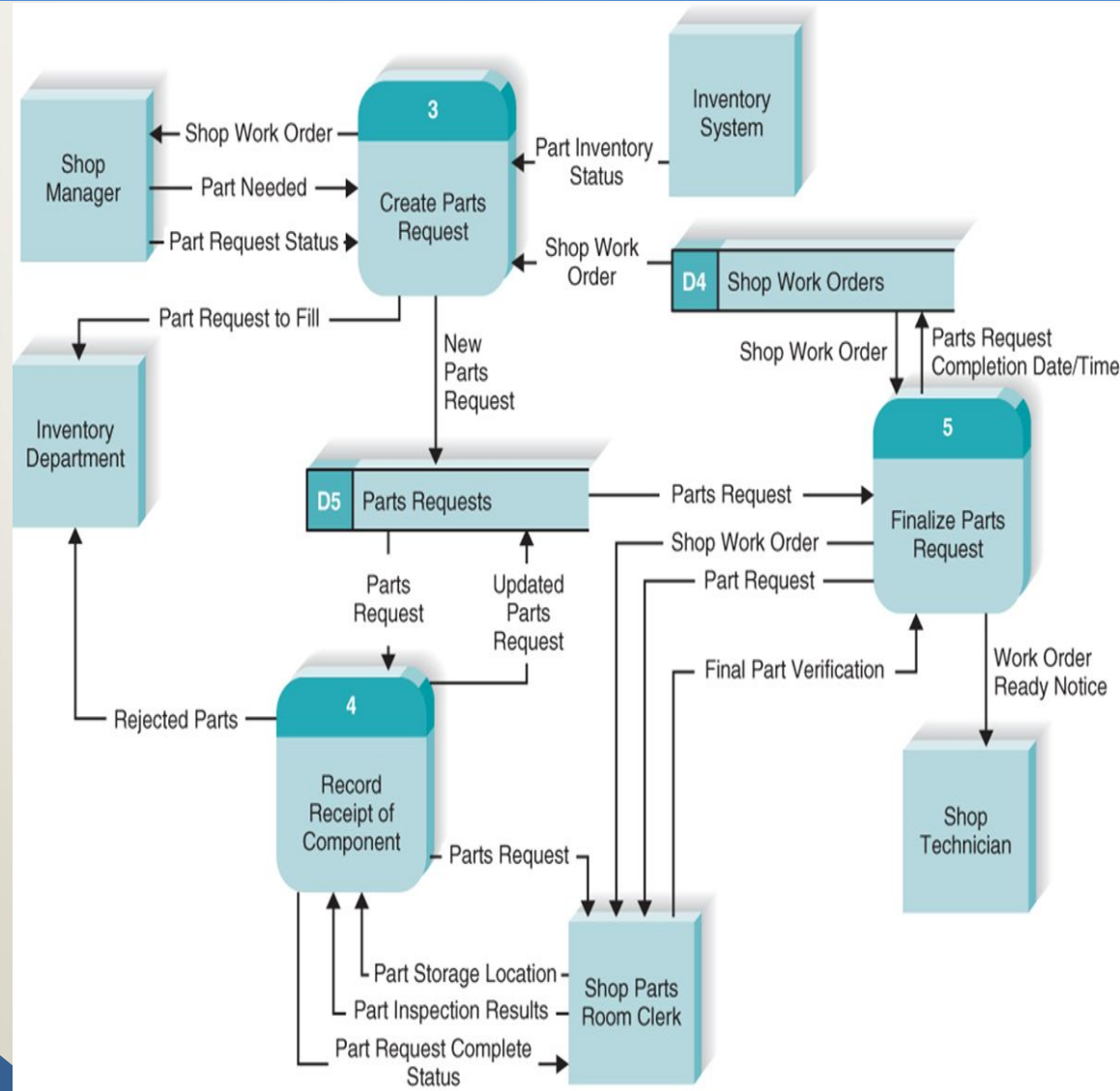


Create a DFD “Fragment” Based on a Use Case

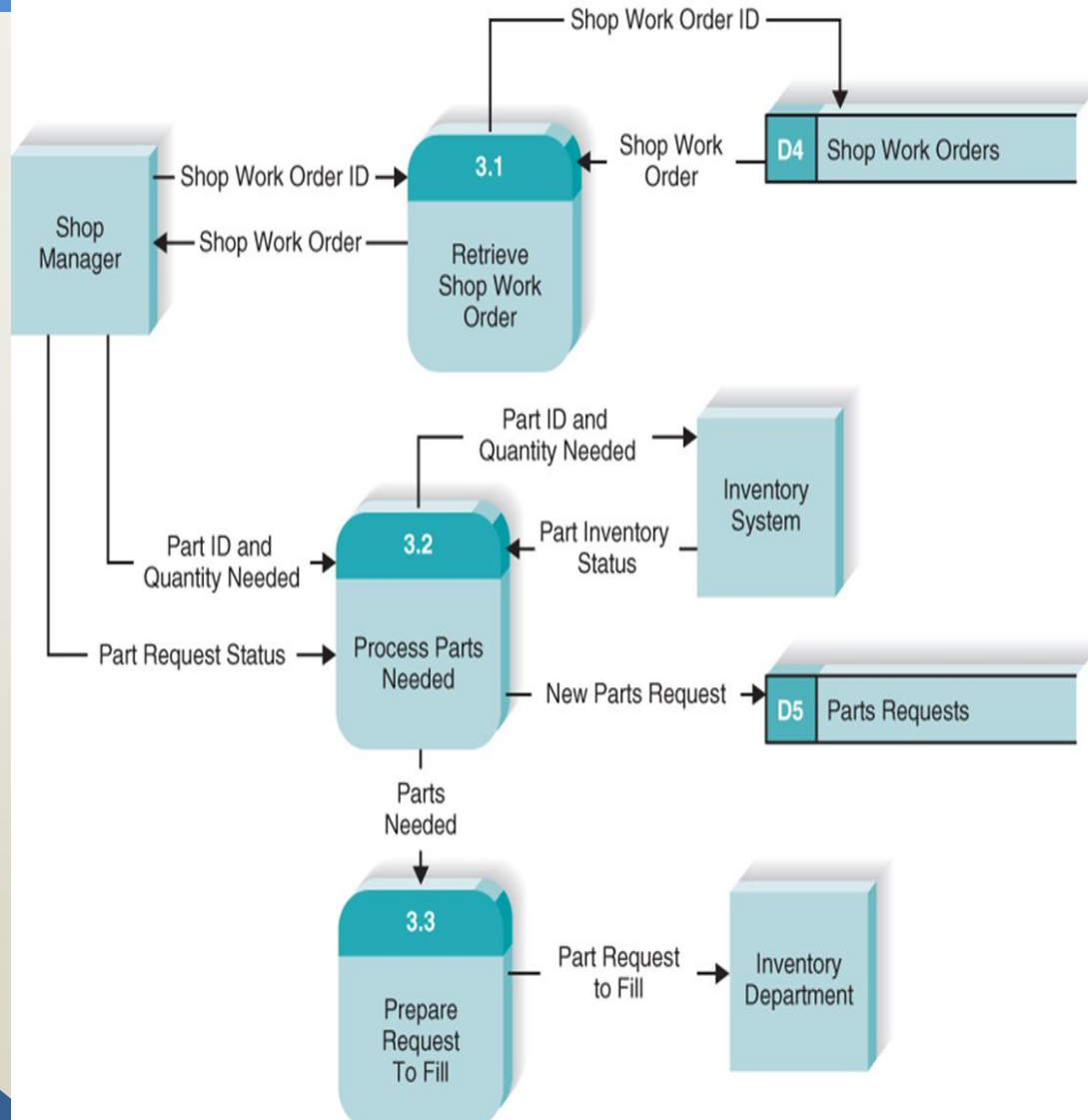
Use Case Name: Create Parts Request		ID: UC-3	Priority: High
Actor: Shop Manager			
Description: This use case describes how the Shop Manager creates a Parts Request			
Trigger: Shop Manager receives notice of new shop work order arrival from Sales System			
Type: ☒ External ☐ Temporal			
Summary Inputs	Source	Summary Outputs	Destination
Part ID and quantity Part inventory status Part request status Shop work order	Shop manager Inventory System Shop manager Shop work order data store	Shop work order New part request record Parts Request to fill	Shop manager Parts Request datastore Inventory department



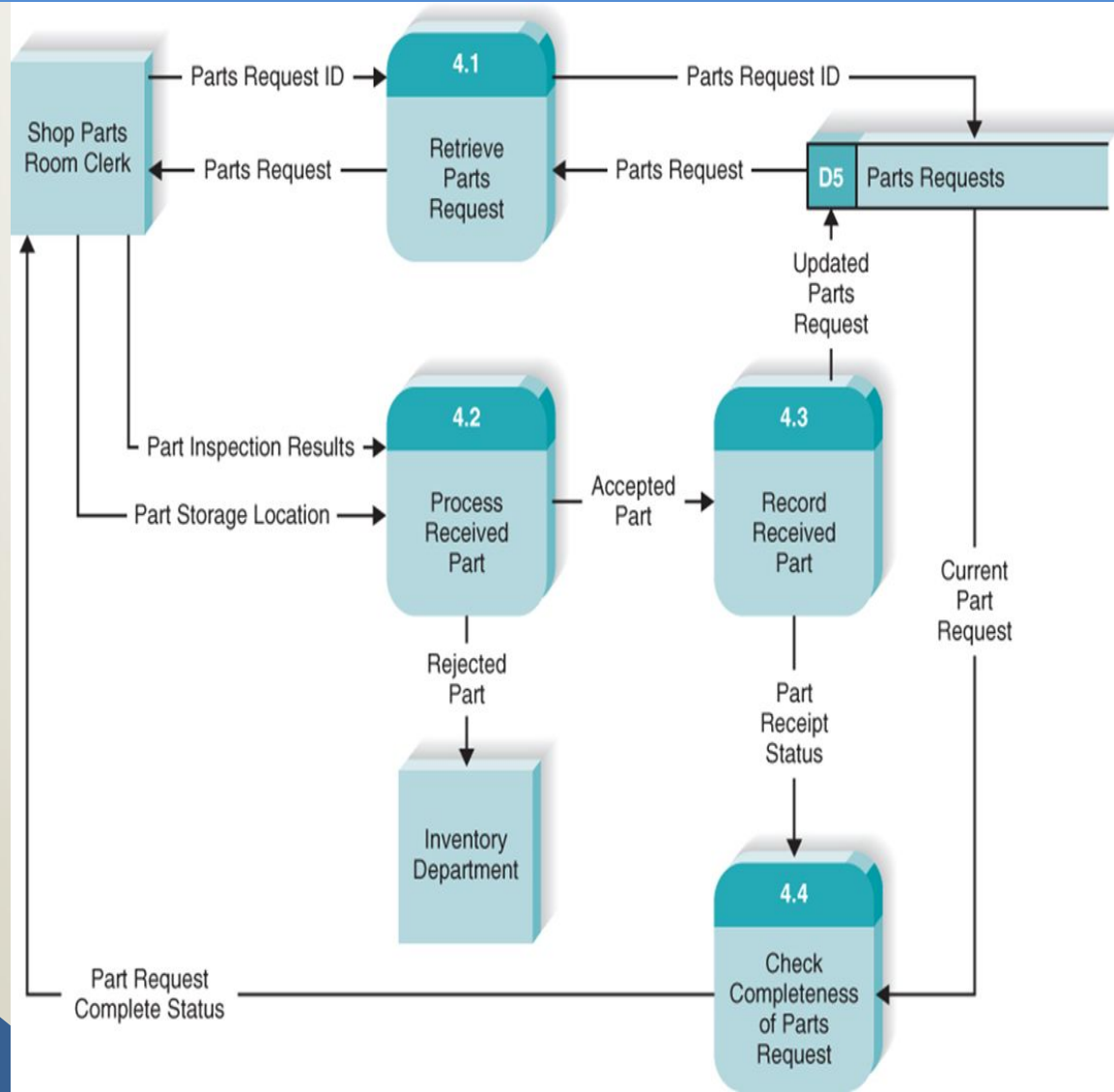
Merge DFD “Fragment” Diagrams into the Level 0 Diagram



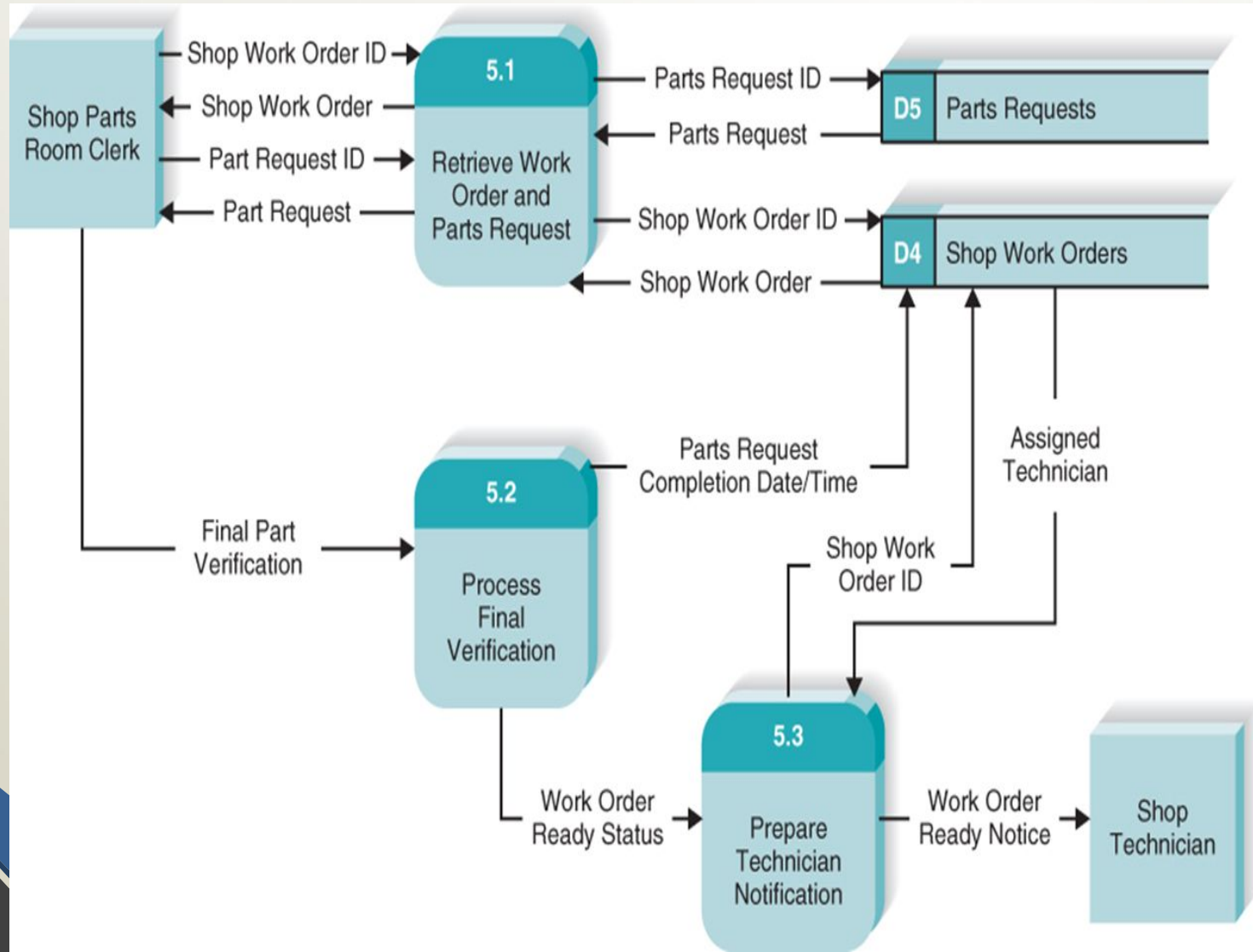
Level 1 – Process 3: Create Parts Request



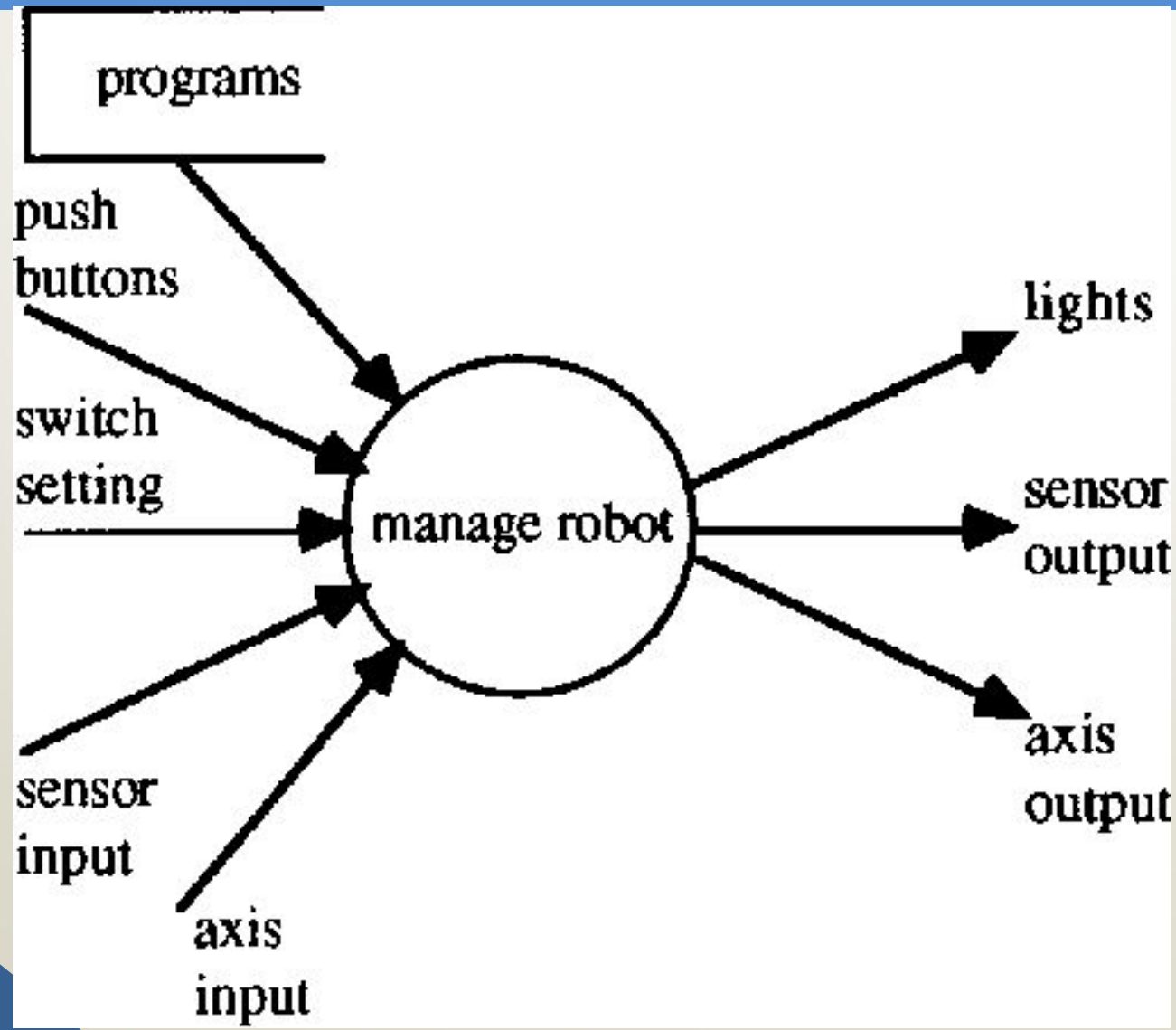
Level 1 – Process 4: Record Receipt of Components



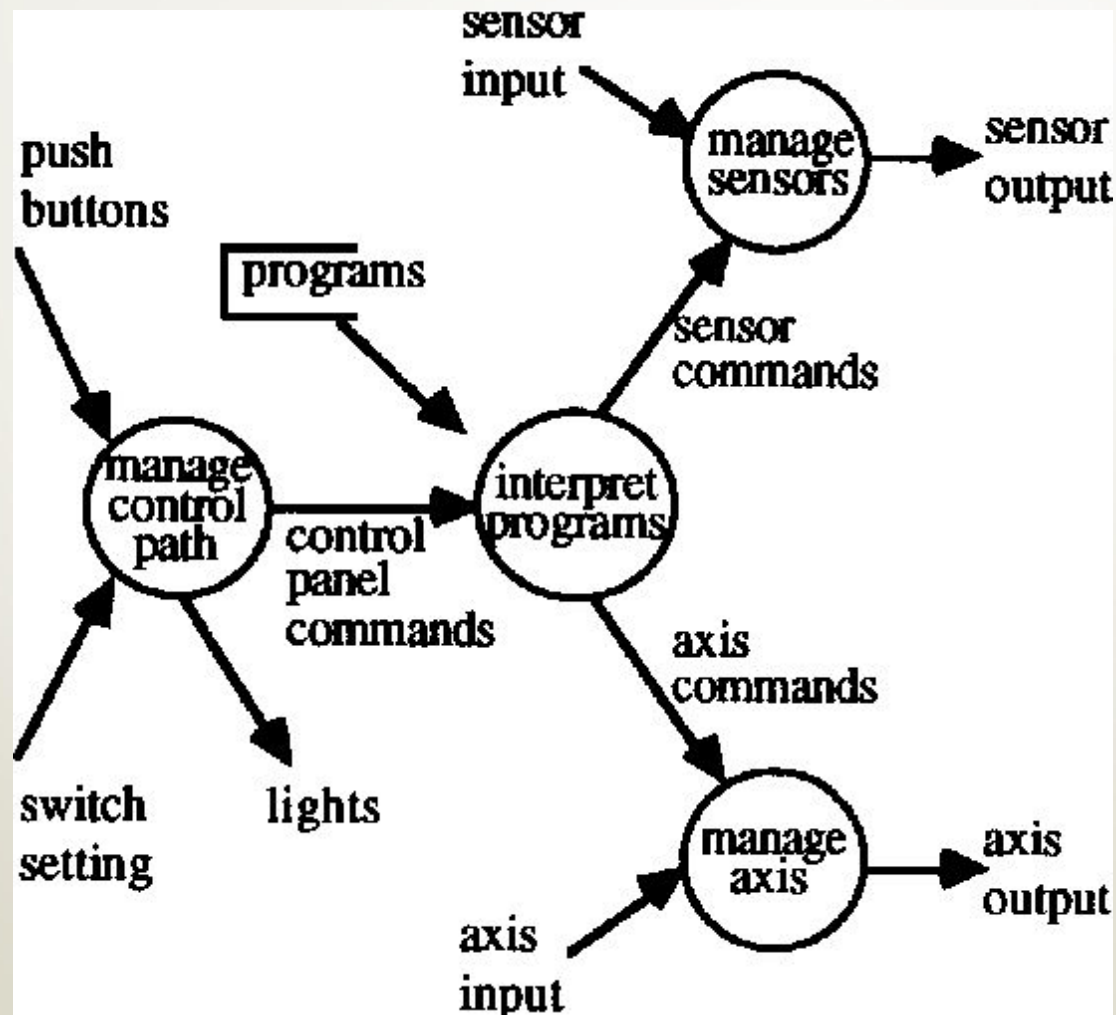
Level 1 – Process 5: Finalize Parts Request

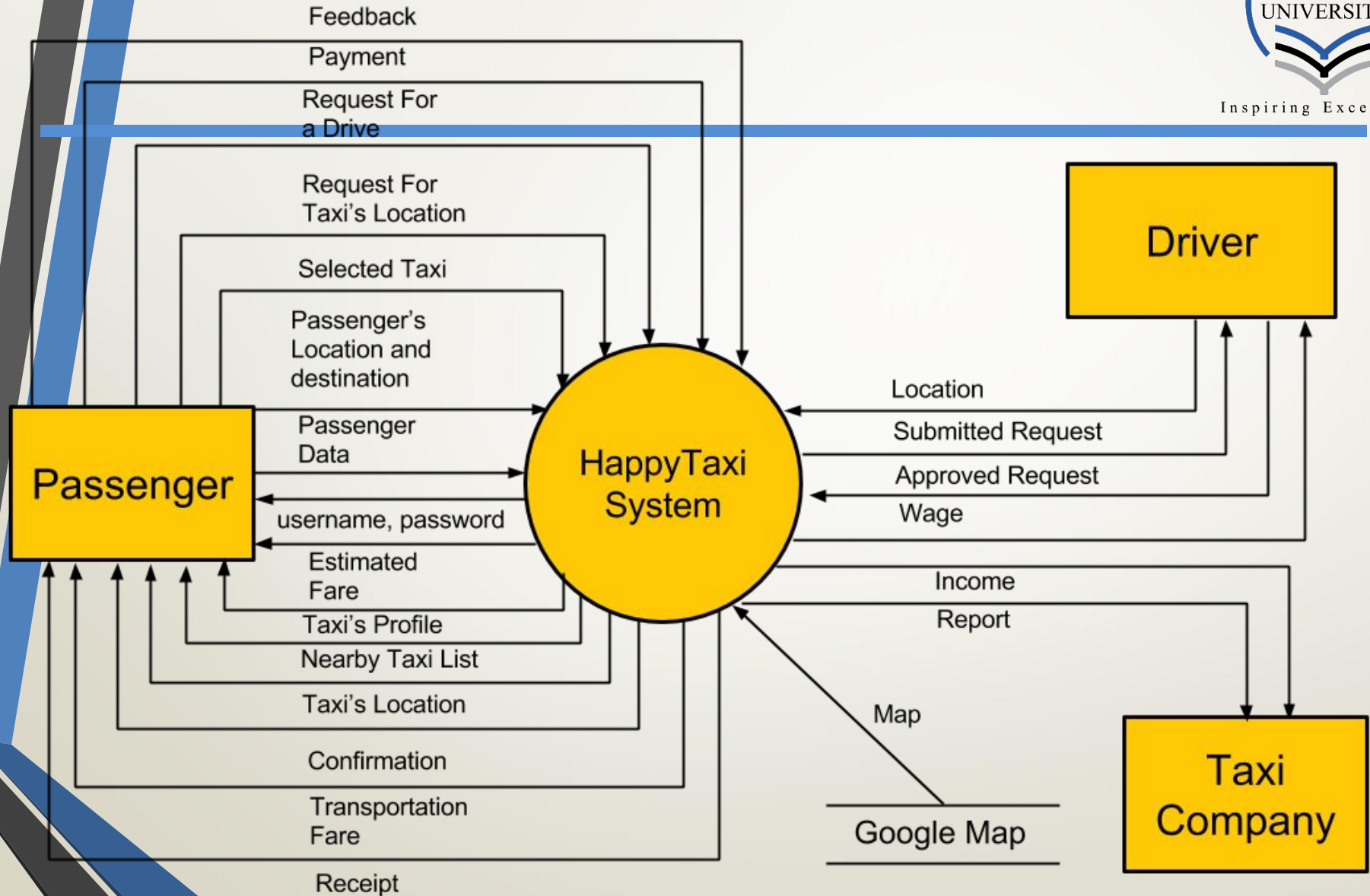


Manage Robot

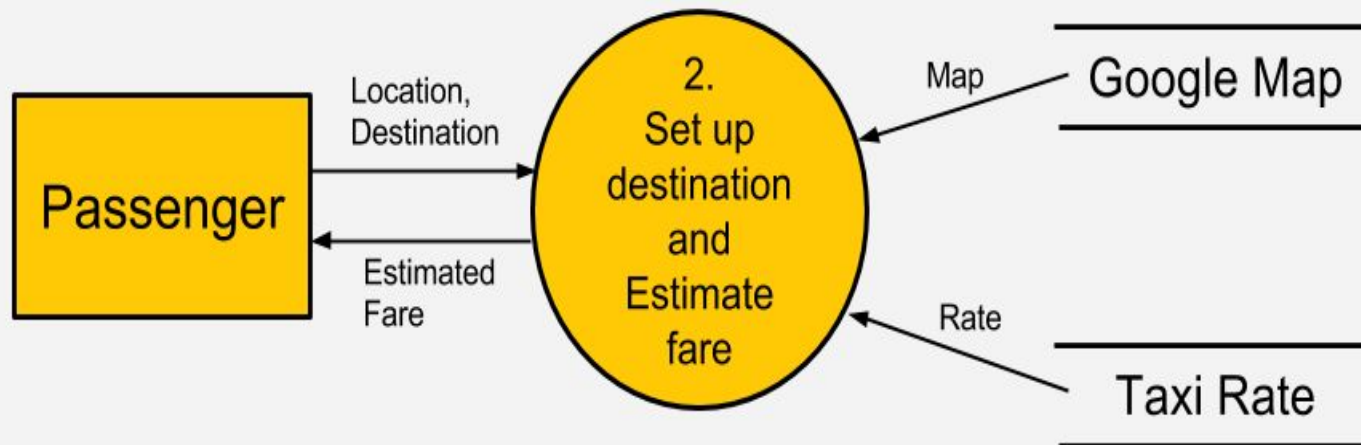


Manage Robot decomposed





Here passenger input current location and destination address to set up destination and estimate fare process and the process send estimate fare to the passenger. The same process gather map information from Google map and rate from taxi rate data base.

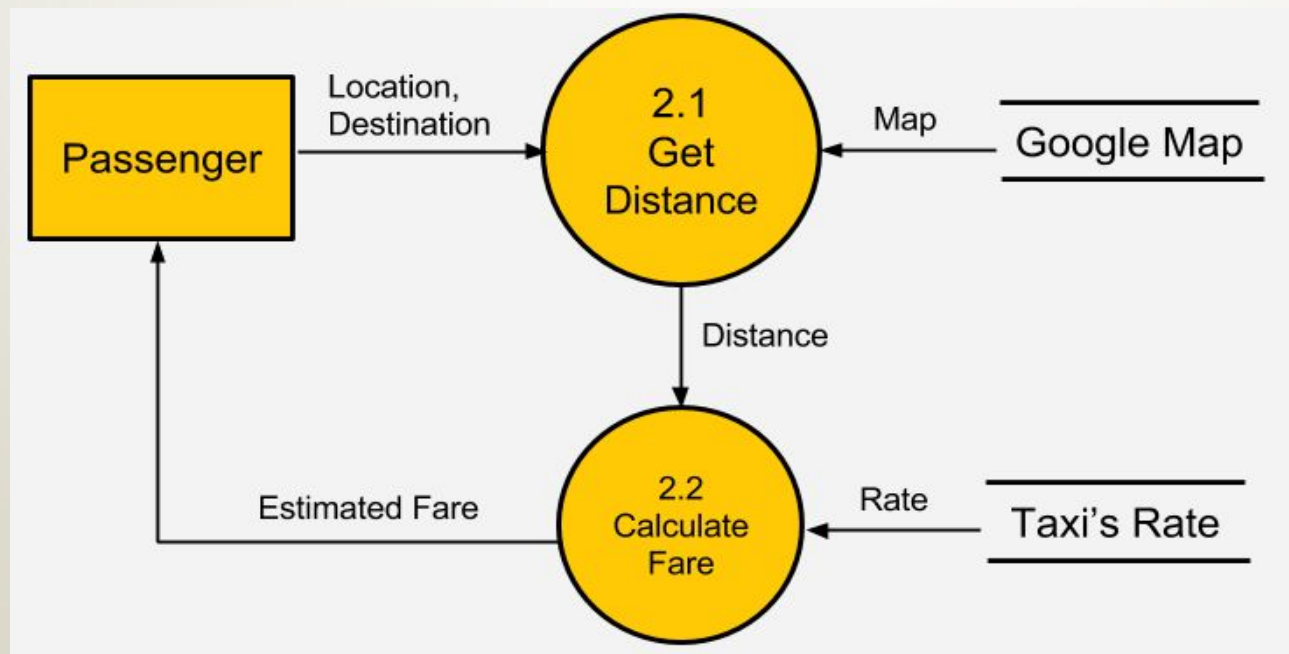


Class work

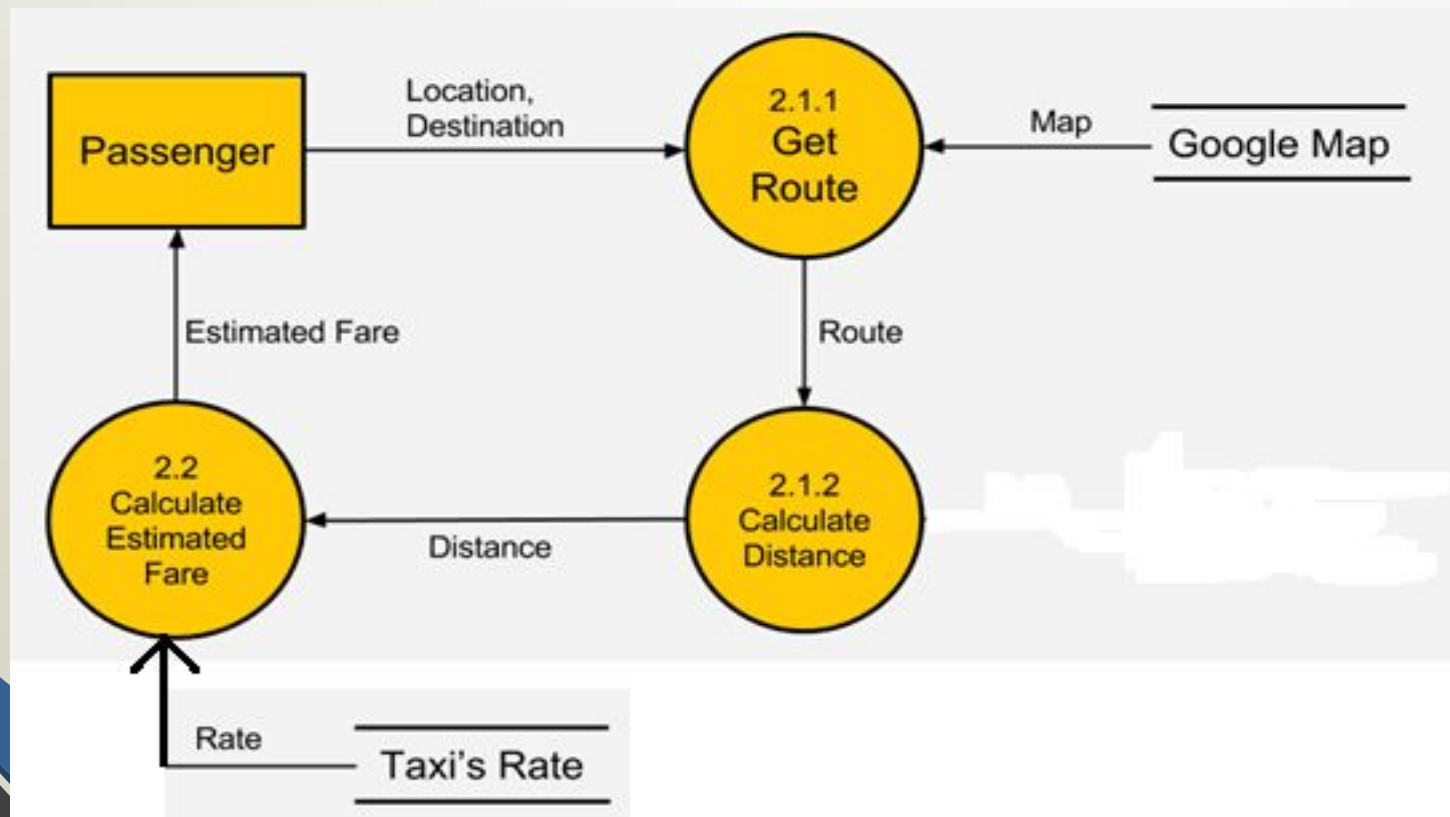
- i. **Draw level 1 DFD** where set up destination and estimate fare process as decomposed into two sub processes as get distance and calculate fare. Data of distance will be sent from get distance to calculate fare.

- i. **Draw Level 2 DFD** for get distance sub process. Decompose get distance in two sub processes as get route and calculate distance. Data of route will sent from get route to calculate distance.

- i. **Draw level 1 DFD** where set up destination and estimate fare process is decomposed into two sub process as get distance and calculate fare. Data of distance will sent from get distance to calculate fare.



Draw Level 2 DFD for get distance sub process. Decompose get distance in two sub process as get route and calculate distance. Data of route will sent from get route to calculate distance.



Chapter Review

- Define the meaning and purpose of the four basic symbols found on a data flow diagram.
- Explain the concept of decomposition and why process models are created as a hierarchy of DFDs.
- Describe several common syntax and semantic errors found on DFDs.
- Discuss the process used to create a process model.
- Discuss how the process model contributes to the development of the new information system.